

Xihan Yao 姚希翰

Austin, TX | [✉ xyaoaf@utexas.edu](mailto:xyaoaf@utexas.edu) | [in LinkedIn](#) | [GitHub](#) | [Website](#)

RESEARCH INTERESTS

My research interests center on integrating Remote Sensing, GIS, and GeoAI technologies for multimodal, explicit geospatial and environmental modeling, with a focus on heterogeneous, human–nature coupled landscapes. I am particularly interested in analyzing ecosystem services and examining how diverse landscape features interact to shape environmental and social resilience.

EDUCATION

- 2025 – present **The University of Texas at Austin** — PhD student, Geography and the Environment
Supervisor: Professor Yuhao Kang
Link: [GISense Lab](#)
- 2021 – 2023 **University of California, Berkeley** — MLA, Environmental Planning
Thesis: Towards more effective urban vegetation monitoring and management: Studying of urban tree characteristics and their relationship to the urban heat island effect with high-resolution remote sensing products, a case study in Portland, Oregon, USA.
Link: [Publication](#)
- 2017 – 2021 **Hong Kong University of Science and Technology (HKUST)** — BSc, Environmental Management and Technology
Capstone Project: Light pollution impact assessment in Hong Kong: Multi-dimensional measurement and spatial numerical modelling on integrated light sources in the neighborhood level.
Link: [Publication](#)
- 2019 **University of Washington, Seattle** — Exchange Student

EXPERIENCE

- The University of Texas at Austin — Research Assistant** Austin, TX, 2025 – present
- Leading a preregistered, national-scale research project modeling relationships between built environment characteristics and Big Five personality traits across U.S. urbanized areas.
 - Designing reproducible, Python-based workflows to process and harmonize TB-scale, multi-modal geospatial data, including Google Street View imagery, high-resolution land cover, POI data, and human mobility records.
 - Applying Earth foundation models (e.g., Google AlphaEarth) and statistical modeling to evaluate whether high-capacity embeddings provide additional explanatory power beyond traditional built-environment variables.
 - Conducting model diagnostics and spatial analyses (residual analysis, spatial autocorrelation tests) to identify geographic outliers, systematic failure modes, and limits of model generalizability.
 - Supervising four undergraduate research assistants and coordinating collaborative research across the lab and with external collaborators.
- EarthDefine, LLC — GIS Analyst** Redmond, WA, 2024 – 2025
- Designed and implemented deep learning–based raster classification workflows for city-scale, sub-meter land cover mapping and change detection, emphasizing reproducible batch processing and quality control.
 - Led a team of GIS specialists to deliver operational geospatial data products across the U.S., including community-level canopy coverage metrics and multi-year land-use change datasets; supported external stakeholders such as the Arbor Day Foundation and municipal governments.
- University of California, Berkeley** Berkeley, CA, 2022 – 2023

—Lecturer, GIS

2023

- Served as instructor of record for UC Berkeley’s largest GIS course (GEOG/LDARCH C188; 200 students), delivering lectures, designing assessments, and supervising a multi-TA instructional team.
- Redesigned and modernized the curriculum to incorporate contemporary GIS concepts and tools, working with ESRI to improve data workflows, reproducibility, and instructional data management.

—Graduate Student Instructor, Applied Remote Sensing & GIS

2022 – 2023

- Taught applied remote sensing labs using Google Earth Engine (JavaScript), covering object- and pixel-based image analysis, machine-learning land use classification, and multi-source data integration (Landsat, MODIS, NAIP, LiDAR).
- Led hands-on GIS labs in ArcGIS Pro, updated legacy course materials, and produced instructional videos to support the transition from ArcGIS Desktop to ArcGIS Pro.

—Research Assistant

2022

- *Korean Tidal Flats, Hwaseong Wetland World Heritage Inscription Study* ([Project Report](#)): Conducted spatial analysis of environmental and social factors to illuminate the significance and challenges of its World Heritage site designation.

Beijing Smart & Green Transport Technology — Data Research and Analysis Intern

2020

- Analyzed carbon abatement potential in China’s transportation sectors under various green technology incentive policy scenarios; utilized a multi-parameter model to assess effectiveness under the supervision of Dr. Dongquan He.

PUBLICATIONS

3. **Yao, X.**, Kim, M., Dronova, I., McBride, J. R., Kondolf, G. M., & Radke, J. D. (2025). Community-scale microclimate simulation using Airborne Laser Scanning and object-based urban tree classification. *Landscape and Urban Planning*, 263, 105420. DOI: [10.1016/j.landurbplan.2025.105420](https://doi.org/10.1016/j.landurbplan.2025.105420)
2. Tong, J. C. K., **Yao, X. H.**, Lau, E. S. L., Cheung, W. K. S., Ho, K. K. S., Ng, V. Y. Y., & Lau, A. P. S. (2023). Light pollution impact assessment in Hong Kong: Multi-dimensional measurement and spatial numerical modelling on integrated light sources in the neighbourhood level. *Energy & Environment*, 35(5), 2497–2516. DOI: [10.1177/0958305X221146942](https://doi.org/10.1177/0958305X221146942)
1. Chan, S. N., Fan, Y. W., & **Yao, X. H.** (2022). Mapping of coastal surface chlorophyll-a concentration by multispectral reflectance measurement from unmanned aerial vehicles. *Journal of Hydro-environment Research*, 44, 88–101. DOI: [10.1016/j.jher.2022.08.003](https://doi.org/10.1016/j.jher.2022.08.003)

CONFERENCE PRESENTATIONS & PROCEEDINGS

6. **Yao, X.**, Kim, M., Dronova, I., McBride, J. R., Kondolf, G. M., & Radke, J. D. (2025, October 23–25). *Community-scale microclimate simulation using Airborne Laser Scanning and object-based urban tree classification* [Paper presentation]. ACSP 2025 Minneapolis 65th Annual Conference, Minneapolis, MN, United States.
5. Main, K. L., **Yao, X.**, Raine, H., Farley, M., Malone, H., Serra-Llobet, A., Kondolf, G. M., & Radke, J. (2024, June 23–27). *The levee effect and residual risk in Sacramento, California: Have we been good or lucky?* [Paper presentation]. ASFPM Foundation 14th Annual Student Paper Competition, Salt Lake City, UT, United States.
4. **Yao, X.**, & Kim, M. (2024, June 3–7). *Object-based ecosystem services evaluation with higher spatial res-*

olution, multidimensional remote sensing [Poster presentation]. 2024 Natural Capital Symposium, Stanford, CA, United States.

3. **Yao, X., & Kim, M.** (2023, December 11–15). *A Lidar-based method for 3D urban forest evaluation and microclimate assessment, a case study in Portland, Oregon, USA* [Poster presentation]. American Geophysical Union (AGU) Annual Meeting 2023, San Francisco, CA, United States. DOI: [10.22541/essoar.170914530.09781933](https://doi.org/10.22541/essoar.170914530.09781933)
2. **Yao, X., & Kim, M.** (2023, July 16–21). *From local to micro: Exploratory data analysis on urban forests and microclimates in Portland, Oregon, USA* [Conference session]. IGARSS 2023 - 2023 IEEE International Geoscience and Remote Sensing Symposium, Pasadena, CA, United States. DOI: [10.1109/IGARSS52108.2023.1028208](https://doi.org/10.1109/IGARSS52108.2023.1028208)
1. Hester, R., McNally, M., Stern, M., **Yao, X.**, & Malone, H. (2022, September 8). *The factor of time, Hwaseong Wetland* [Conference session]. Association of Pacific Rim Universities Sustainable Cities and Landscapes (APRU-SCL), Honolulu, HI, United States (Remote).

AWARDS & CERTIFICATES

Honors and Awards

- **Energy AI Hackathon (6th Annual)**, UT Austin —Hildebrand Department of Petroleum and Geosystems Engineering (PGE).
Awarded **Fourth Place** in a competitive machine learning challenge (46 teams). Developed a "similarity–geospatial proximity gated model" achieving strong predictive accuracy and interpretability for subsurface production forecasting under sparse, heterogeneous conditions. 2026
- **Geoscience Hackathon**, UT Austin —Jackson School of Geosciences.
Awarded **Second Place**. Project: "Tracking Earth's Changes with AlphaEarth Foundations," applying foundation model embeddings for large-scale, flexible landscape system change detection. 2025
- **CyberTraining Summer School**, Texas A&M University —Department of Geography.
Awarded **First Place**. Project: "Assessing Instance-based Disaster Impact Through Multimodal Geospatial Data" 2025
- **Collegiate Student Paper Competition**, Association of State Floodplain Managers (ASFPM).
Awarded **Third Place**. Paper: "The Levee Effect and Residual Risk in Sacramento, California" 2024
- **IEEE GRSS GeoPitch Award**, IEEE Geoscience and Remote Sensing Society.
Awarded **Third Place**. Elevator Pitch Presentation: "Object-Based Urban Trees Characterization with Airborne LiDAR" 2023

Professional Certificates

- Certificate in Geographic Information Science and Technology (GIST), UC Berkeley. 2023
- Certificate of Teaching and Learning in Higher Education, UC Berkeley 2023

PEER REVIEW SERVICE

- *Urban Forestry & Urban Greening* (Impact Factor: 6.7)
- *Journal of Applied Remote Sensing* (Impact Factor: 1.4)
- *Computational Urban Science* (Impact Factor: 3.2)

Mentorship & Leadership

- **NASA DEVELOP National Program** Fall 2024
Technical Mentor for Student Internship Project: “Quantifying Canopy Cover and Land Surface Temperature in San Jose to Identify Future Tree Planting Sites”.

- **Undergraduate Honors Thesis Mentor** 2023 – 2024
Thesis Mentor for Sarah Song (UC Berkeley, Statistics & Data Science). Project: “Urban Greenery in San Francisco.

Skills

Programming & Data Science	Python (TensorFlow, PyTorch, Scikit-learn, GeoPandas, Rasterio, PySAL), R, JavaScript (Google Earth Engine)
GIS & Remote Sensing	ESRI products (ArcGIS Pro, ArcGIS Online), QGIS, Google Earth Engine, Google AlphaEarth Foundations, LiDAR
Computing & Systems	Linux Shell, High-Performance Computing (HPC), GPU server management, batch processing, Git/GitHub